

Chemical Process Calculations Roadmap

CHE251

1 Fundamentals of Process Calculations

1.1 Process Variables and System Description

- Basis, mass vs. molar quantities, flow rates, density, mass fraction x_i , mole fraction y_i , average molecular weight.
- Continuous, batch and semi-batch processes; open vs. closed systems.
- Steady state (no accumulation, all variables constant with time) vs. unsteady/transient state.

1.2 The General Balance Equation

For any conserved (or reacting) quantity within a system boundary:

$$\text{Accumulation} = \text{Input} - \text{Output} + \text{Generation} - \text{Consumption} \quad (1)$$

At steady state, accumulation = 0, so:

$$\text{Input} + \text{Generation} = \text{Output} + \text{Consumption} \quad (2)$$

For total mass, and for any species in a non-reactive process, generation = consumption = 0, giving input = output.

2 Material Balances

2.1 Degree-of-Freedom (DOF) Analysis

$$\text{DOF} = (\text{number of unknown variables}) - (\text{number of independent equations}) \quad (3)$$

A problem is solvable when $\text{DOF} = 0$. Independent equations arise from balances (at most one per independent species, or one total + $(n - 1)$ component balances for n species), specified relations (ratios, given compositions), and auxiliary relations (e.g. phase-equilibrium or density relations). Balances on a unit, a group of units, or the overall process boundary may each be used, but only a limited number of them are independent.

2.2 Multiple-Unit Systems

For a flowsheet with several interconnected units, balances may be written around: (i) each individual unit, (ii) the overall process, or (iii) any envelope enclosing two or more units. Choosing the envelope that eliminates unwanted internal streams simplifies the solution.

2.3 Recycle, Bypass and Purge

- **Recycle:** a fraction of an output stream is returned to the process inlet to improve raw-material utilisation.
- **Bypass:** a fraction of the feed skips a unit entirely and rejoins the process downstream.
- **Purge:** a small stream is continuously withdrawn from a recycle loop to prevent the build-up of inert or non-reacting species that would otherwise accumulate indefinitely.
- **Overall conversion** is based on fresh feed to the process; **single-pass conversion** is based on feed entering the reactor (fresh + recycle) for a single pass.

3 Reactive Systems

3.1 Stoichiometry and Extent of Reaction

- **Limiting reactant:** the reactant present in the smallest stoichiometric proportion; all others are in **excess**, quantified as $\% \text{ excess} = \frac{n_{\text{fed}} - n_{\text{stoich}}}{n_{\text{stoich}}} \times 100$.
- **Extent of reaction** ξ relates the moles of each species to stoichiometric coefficients ν_i (negative for reactants, positive for products):

$$n_i = n_{i,0} + \nu_i \xi \quad (4)$$

- **Fractional conversion** = $\frac{\text{reactant reacted}}{\text{reactant fed}}$.

3.2 Atomic Balances

Atomic species are conserved even when molecular species react, so for every atomic element:

$$\text{atoms in} = \text{atoms out} \quad (5)$$

Atomic balances are often simpler than molecular balances for systems with multiple or unknown reactions, and together with an energy balance (where relevant) they may be sufficient to close the DOF without knowing the exact reaction stoichiometry.

3.3 Yield and Selectivity

For processes with side reactions:

- **Yield** = $\frac{\text{moles of desired product formed}}{\text{moles that would form if the limiting reactant were fully converted with no side reactions}}$.
- **Selectivity** = $\frac{\text{moles of desired product}}{\text{moles of undesired product}}$.

4 Combustion

- **Theoretical (stoichiometric) air:** the amount of air that supplies exactly the O_2 needed for complete combustion; **excess air** is any amount supplied beyond this.
- **Complete combustion** converts all C to CO_2 and H to H_2O ; **incomplete combustion** produces CO (and possibly soot) as well.
- **Flue (stack) gas** composition may be reported on a wet basis (including water vapour) or a dry basis (water condensed out, as in an Orsat analysis).
- **Standard heat of combustion** $\Delta \hat{H}_c^0$ is the enthalpy change of the standard reaction of a substance with O_2 to form CO_2 , H_2O and other standard combustion products (e.g. SO_2). It equals the **higher heating value (HHV)** when H_2O is a liquid product, and the **lower heating value (LHV)** when H_2O is a vapour product; the two differ by the latent heat of vaporisation of the water formed.

5 Vapor–Liquid Equilibrium

- **Raoult's Law** (ideal solution, ideal vapour): $y_i P = x_i p_i^*(T)$, where $p_i^*(T)$ is the pure-component vapour pressure.
- **Antoine equation:** $\log_{10} p^* = A - \frac{B}{T + C}$, used to compute vapour pressure at a given temperature.

- **Bubble point:** the temperature (at fixed P) at which a liquid mixture first begins to vaporise, found by solving $\sum_i x_i p_i^*(T) = P$.
- **Dew point:** the temperature at which a vapour mixture first begins to condense, found from $\sum_i y_i P / p_i^*(T) = 1$.
- For a system of C components and P phases in equilibrium, the Gibbs phase rule gives the degrees of freedom as $F = C - P + 2$; specifying this many intensive variables (e.g. T and the liquid-phase mole fractions) fixes the remaining phase compositions.
- **Trouton's Rule:** $\frac{\Delta \hat{H}_v(T_b)}{T_b} \approx \text{constant}$ ($\sim 88 \text{ J/mol K}$ for many non-polar liquids), used to estimate the heat of vaporisation at the normal boiling point when data are unavailable.

6 Energy Balances

6.1 General Energy Balance

For an open, steady-state system, neglecting kinetic and potential energy changes:

$$\Delta \dot{H} = \dot{Q} - \dot{W}_s \quad (6)$$

where $\dot{Q}$ is the rate of heat added to the system and $\dot{W}_s$ is the shaft work done by the system.

6.2 Enthalpy Changes: Sensible and Latent Heat

- **Sensible heat** (temperature change, no phase change): $\Delta \hat{H} = \int_{T_1}^{T_2} C_p dT$.
- **Latent heat** (phase change at constant T): heats of vaporisation, fusion and sublimation.
- Enthalpy changes are path-independent; a convenient hypothetical path (e.g. heat $\rightarrow$ vaporise $\rightarrow$ heat) may be used between any two states.

6.3 Heats of Reaction and Combustion

The standard heat of reaction can be computed from standard heats of formation ($\Delta \hat{H}_{f,i}^0$) via Hess's Law:

$$\Delta \hat{H}_r^0 = \sum_{\text{products}} \nu_i \Delta \hat{H}_{f,i}^0 - \sum_{\text{reactants}} \nu_i \Delta \hat{H}_{f,i}^0 \quad (7)$$

Alternatively, heats of reaction may be obtained from standard heats of combustion of reactants and products. A reaction is **exothermic** if $\Delta \hat{H}_r^0 < 0$ and **endothermic** if $\Delta \hat{H}_r^0 > 0$.

6.4 Energy Balances on Reactive Processes

- **Heat-of-reaction method:** reference states are elemental species; ΔH is built from $\Delta \hat{H}_r^0$ plus sensible/latent heat terms to bring streams to reaction conditions.
- **Adiabatic reactor** ($\dot{Q} = 0$): all the heat released (or absorbed) by reaction goes into changing the temperature of the product stream, leading to the **adiabatic flame/reaction temperature**.
- **Isothermal exothermic reactor:** heat must be *removed* to keep temperature constant, so $\dot{Q} < 0$; for an isothermal endothermic reactor, $\dot{Q} > 0$.

7 Psychrometry

- **Dry-bulb temperature:** the temperature read by an ordinary thermometer exposed to humid air.

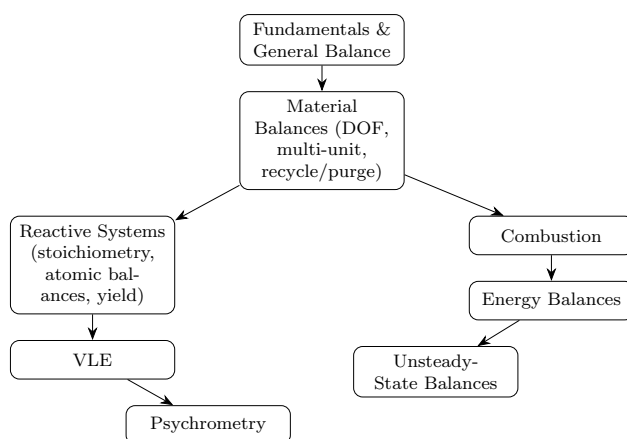
- **Wet-bulb temperature:** the steady-state temperature reached by a wetted thermometer bulb in flowing air, governed by simultaneous heat and mass transfer.
- The wet-bulb temperature equals the dry-bulb temperature only when the air is saturated (100% relative humidity); otherwise the wet-bulb reading is lower.

8 Unsteady-State Balances

When accumulation cannot be neglected, the balance is a differential equation in time:

$$\frac{dM}{dt} = \dot{m}_{\text{in}} - \dot{m}_{\text{out}} \quad (8)$$

Typical problems (e.g. a stirred tank filling, emptying, or approaching a new steady state) require integrating this equation, often assuming constant density so that volume replaces mass.



9 Laboratory Component

9.1 Important Lab Topics

- Steady-state material balances on single- and multiple-unit processes, including systems with recycle and purge, are among the more demanding lab topics and warrant particular preparation.
- DWSIM-based process simulation labs form a distinct and scoring-friendly component of the lab component.
- Equation solving in MATLAB (linear and nonlinear systems, and where relevant, ODEs) is the primary computational task across most labs.

9.2 General Tips

- Review the topic scheduled for each lab beforehand; seek help from teaching assistants during the session, and attend feedback sessions to identify where marks were lost.
- Become comfortable with MATLAB syntax and built-in functions early, particularly in the first two to three labs, since solving equations is the primary task in nearly all subsequent labs.
- For labs, ensure familiarity with the basics of ESC113 and MATLAB: defining functions and variables, and the basic syntax of ODE solvers, `fsolve`, and similar routines.
- Practise formulating mass balances for multiple-unit processes with recycle and purge before the corresponding lab sessions.
- Do not miss the DWSIM simulation labs, as they represent a good opportunity to score marks.

- Allocate time during the lab to the presentation of the laboratory report and to commenting code, as both contribute to the score.
- The mid-semester and end-semester examinations each include a lab-based component, typically a problem similar in style to the lab exercises. Misreading the problem statement under time pressure is a common source of error, so read carefully and remain composed.